OTHER IMPORTANT ADDITION REACTIONS OF ALKENES



▲ Figure 25.5 Production of margarine

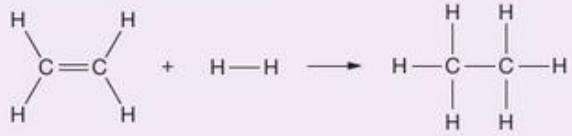
EXTENSION WORK

The addition of water to propene could produce two different products, propan-1-ol and propan-2-ol. In reality, much more propan-2-ol is formed than propan-1-ol. In the major reaction, a hydrogen atom from water is first added onto the end carbon and then the –OH group is attached to the middle carbon. The addition of compounds to an unsymmetrical alkene like propene will be explored in detail when you study chemistry at a more advanced level.

EXTENSION WORK

Hydrogenation

A **hydrogenation** reaction involves the *addition of hydrogen*. An alkene reacts with hydrogen to form an alkane, for example:



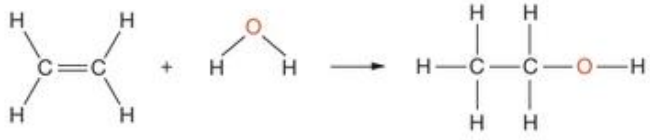
$CH_2{=}CH_2 + H_2 \rightarrow CH_3CH_3$

This reaction is used for the production of margarine, which is made by hydrogenation of polyunsaturated (i.e. many C=C bonds) plant oils.

CHEMISTRY ONLY

HYDRATION

A **hydration** reaction involves the *addition of water*. An alkene reacts with water to form an alcohol, for example:



$CH_2{=}CH_2 + H_2O \rightarrow CH_3CH_2OH$

Starting materials:	ethene and steam
Temperature:	300 °C
Pressure:	60–70 atmospheres (the atmosphere is a unit of pressure, equal to 101325 pascals)
Catalyst:	phosphoric acid

This reaction is used for the production of high-purity ethanol in industry (see Chapter 26 for more detail on this reaction).

END OF CHEMISTRY ONLY

CHAPTER QUESTIONS

SKILLS CRITICAL THINKING

1 The alkenes are a *homologous series* of *unsaturated* hydrocarbons.

a i State two characteristics of the members of a *homologous series*.

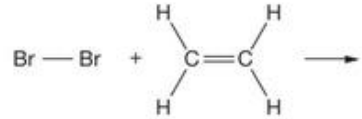
ii Explain the term *unsaturated*.

b Ethene is the simplest member of the alkenes. Bromine water can be used to distinguish alkenes from alkanes.

i Bromine water is added to ethene. State the starting colour of the bromine water and the finishing colour of the reaction mixture.

SKILLS PROBLEM SOLVING, INTERPRETATION

ii Complete the equation by drawing and naming the displayed formula of the product.



Name of the product:

SKILLS INTERPRETATION 8

c Isomers are compounds that have the same molecular formula but different displayed formulae. Draw the displayed formulae of three alkene isomers that have the molecular formula C_4H_8.

d Can you draw any other isomers of C_4H_8 which are *not* alkenes?

SKILLS CRITICAL THINKING 6

2 A gaseous hydrocarbon, **P**, with three carbon atoms, decolourises bromine water.

a Explain what the decolorisation of the bromine water tests for.

7

b Write the displayed formula for the hydrocarbon.

SKILLS PROBLEM SOLVING 10

c Write a balanced chemical equation for the reaction between the hydrocarbon and bromine, using displayed formulae.

d The reaction between ethane and bromine gas in the presence of UV light also causes the bromine to lose colour.

9

i Write a balanced chemical equation for this reaction.

SKILLS REASONING 7

ii This reaction is described as a substitution reaction, whereas the reaction between **P** and bromine is an addition reaction. Using the equations you have already written, explain the difference between addition and substitution.

SKILLS CRITICAL THINKING 8

3 Alkenes undergo addition reactions.

a Predict the structural formula of the compound formed when chlorine reacts with propene. Name the product formed.

b Ethene reacts with hydrogen. Give the name and structural formula of the product.

6

4 Propene can be made by heating propane and sulfur.

a Name another method of making propene and state two conditions for making alkenes from alkanes.

SKILLS EXECUTIVE FUNCTION

b Describe a chemical test which can be used to distinguish between propane and propene.

Test:

Result with propane:

Result with propene:

SKILLS CRITICAL THINKING 8

5 Give the structural formulae and names of the product when but-1-ene reacts with each of the following:

a steam

b hydrogen chloride (HCl)

c bromine.